EMT archetype analysis — findings at a glance

Within single tumours / cell lines, cells trade off a small set of shared programmes; pooled shapes mostly reflect sample identity; hybrids are interior mixtures, not a corner.

QUESTION 1 · Is there a simplex, and are its corners tasks? | **QUESTION 2**

	Pooled over samples A all · B KS · C Hallmark			Within each unit (share of units with a simplex)			What the corners are (within units, all gene sets)					Hybrid cells vs other cells as generalists	Bottom line
	A	B	C	A	B	C	Prolif.	Epi-diff.	Mes / ECM	Metab.	Stress		
Mouse skin SCC tumours	ID	mix	prog	4/8	3/8	3/8						▲ leans interior 58% vs 46% (+10 pts)	Proliferation vs matrix inside tumours; hybrids most interior
Cook TGF-β lines cell lines	ID	ID	ID	1/4	1/4	3/4						● no difference 56% vs 55% (+0 pts)	Pooled shape = cell lines; within lines a simplex mostly only with Hallmark genes
Mouse PDAC libraries	mix	–	–	9/9	3/9	4/9						▲ slight lean 66% vs 62% (+5 pts)	Robust within-library triangles; EMT genes alone weaker
4T1 clones clones	mix	mix	–	1/6	1/6	1/6						● no difference 69% vs 66% (+2 pts)	Almost no structure inside clones
MCF10A / HuMEC samples	mix	mix	ID	6/6	4/6	5/6						▲ slight lean 65% vs 56% (+6 pts)	Clearest trade-off: differentiation / EMT-matrix / proliferation

Pooled:

prog — corners are programmes

mix — some corners = one sample

ID — corners = samples (identity)

– no significant simplex

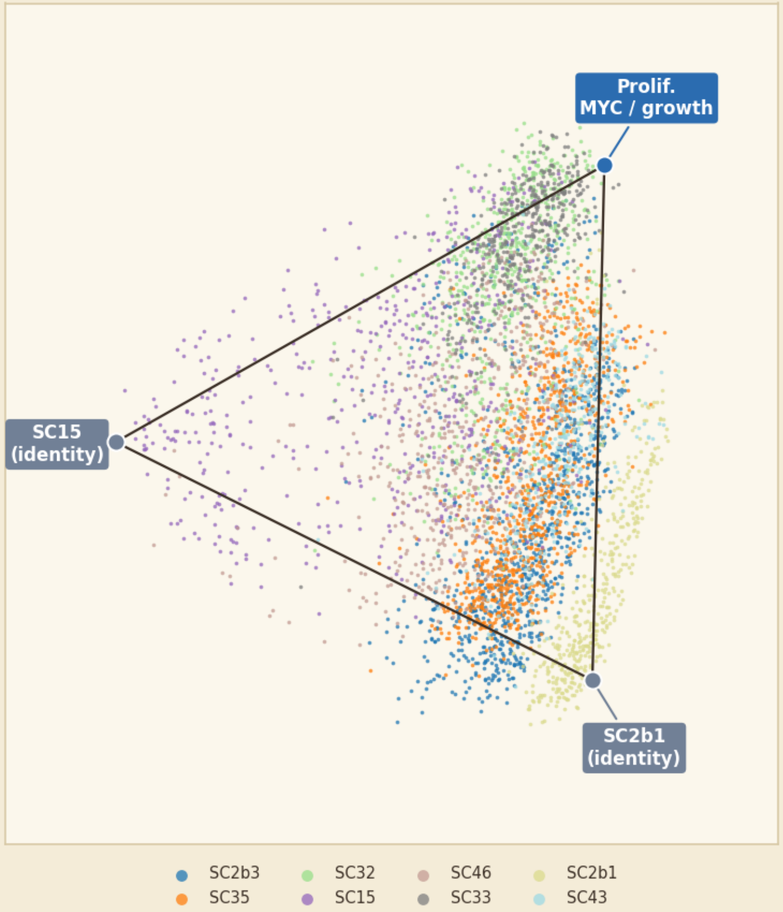
share of units with a significant simplex

Bubble = share of significant within-unit simplices with a corner of that family (faded if < 5 simplices).
Q2: gap rule, median over units with a simplex.

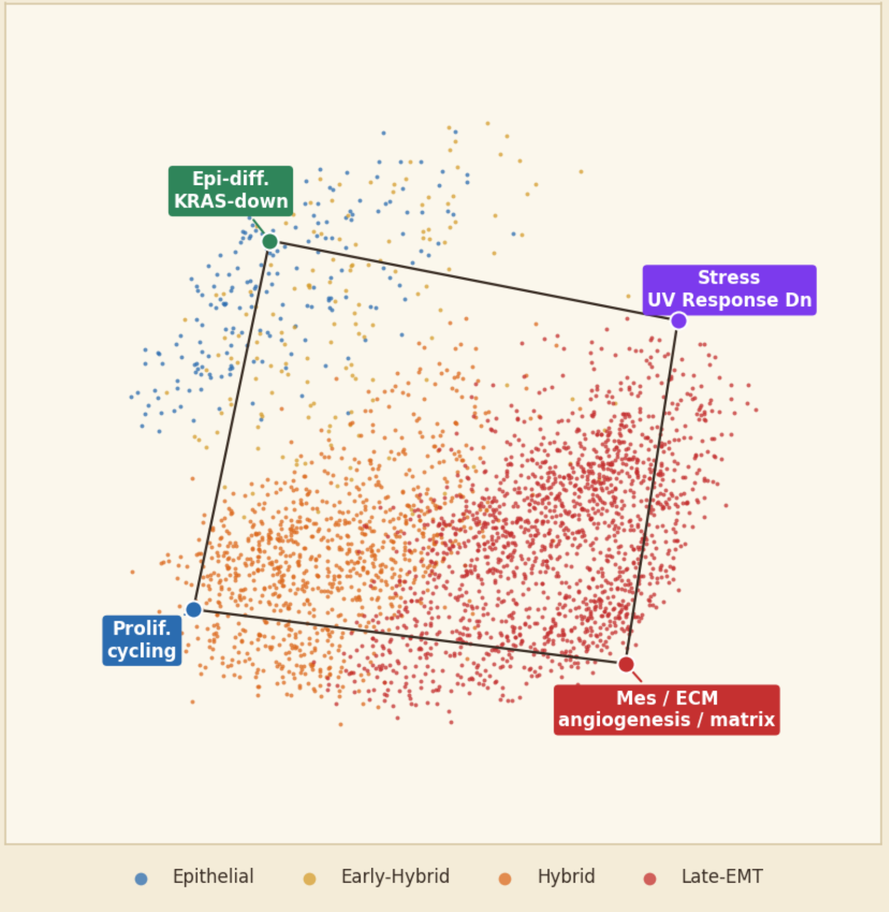
Mouse skin SCC

Inside SCC tumours, cells trade off proliferation against a mesenchymal / matrix programme; hybrid cells sit between the corners

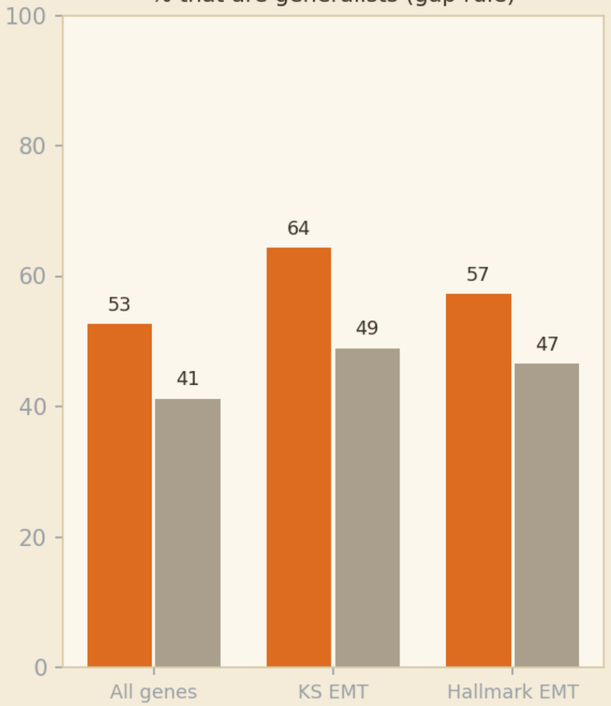
Pooled, all genes: corners mostly = tumours
(cells coloured by tumour; k=3, p=0.005)



Inside one tumour (SC2b3, All genes): corners = programmes
(cells coloured by EMT state; k=4, p=0.005; 4 of 8 tumours have a simplex)



Hybrid cells (orange) vs other cells (grey):
% that are generalists (gap rule)



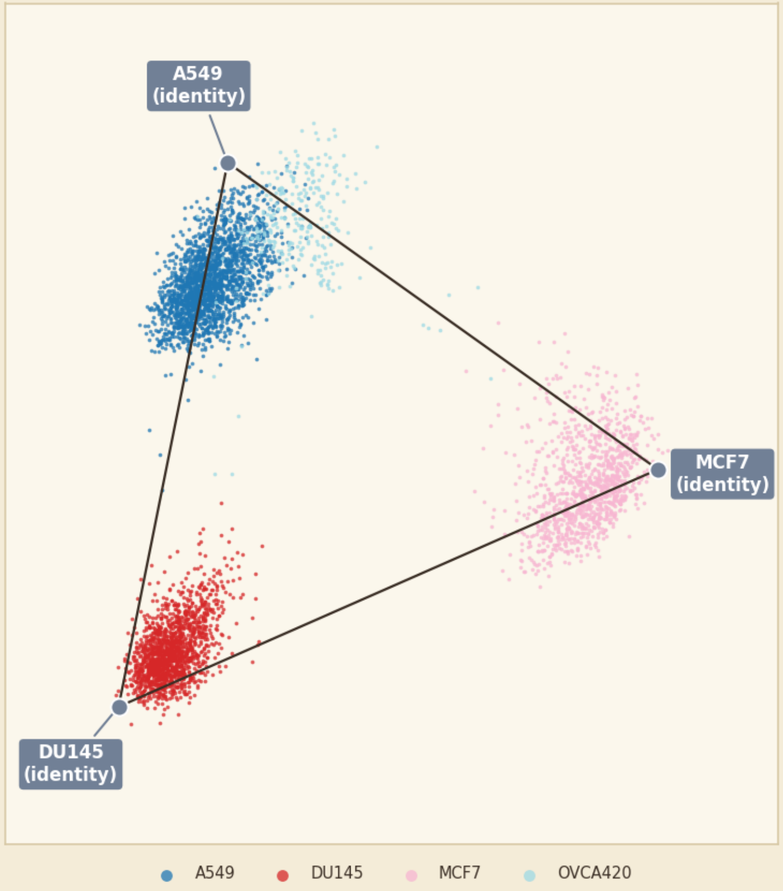
Hybrids lean toward the interior (median +10 points)

Significant within-tumour simplices: all genes 4/8, KS 3/8, Hallmark 3/8. Corner families across them: proliferation (8/10), epithelial differentiation (3/10), mesenchymal / matrix (9/10), metabolism & hypoxia (2/10), stress & immune signalling (6/10).

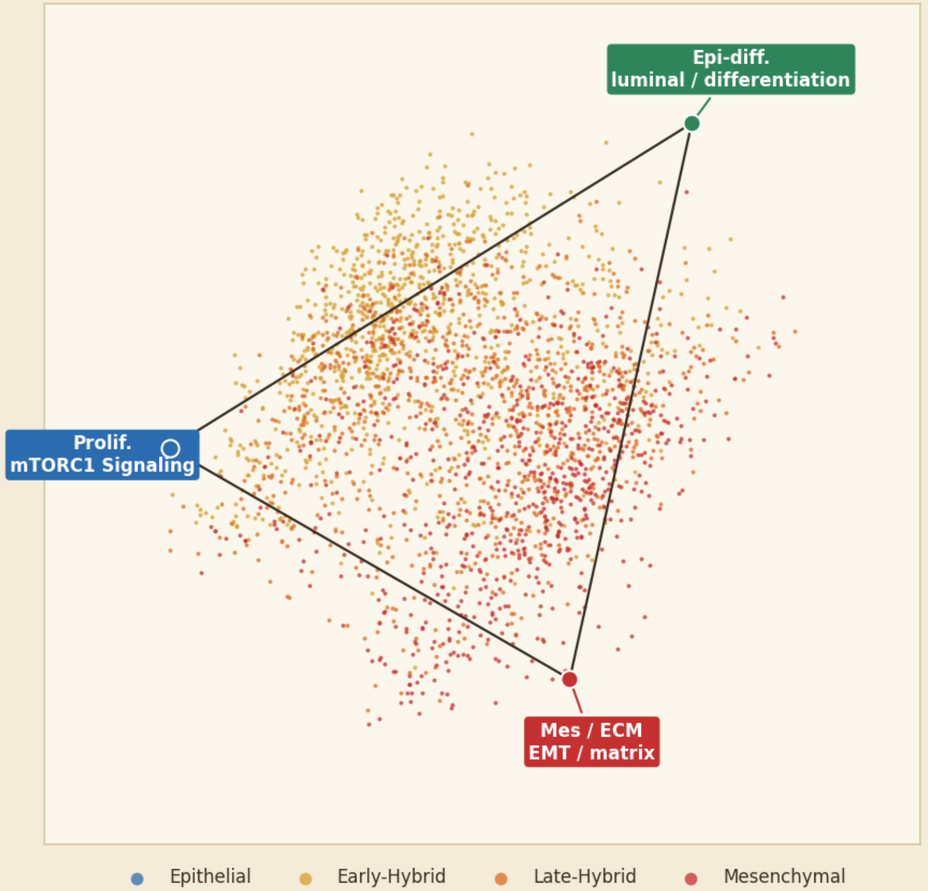
Cook TGF-β lines

In the Cook lines the pooled simplex is just the four cell lines; inside a line a simplex appears mainly with the Hallmark EMT genes (3 of 4 lines)

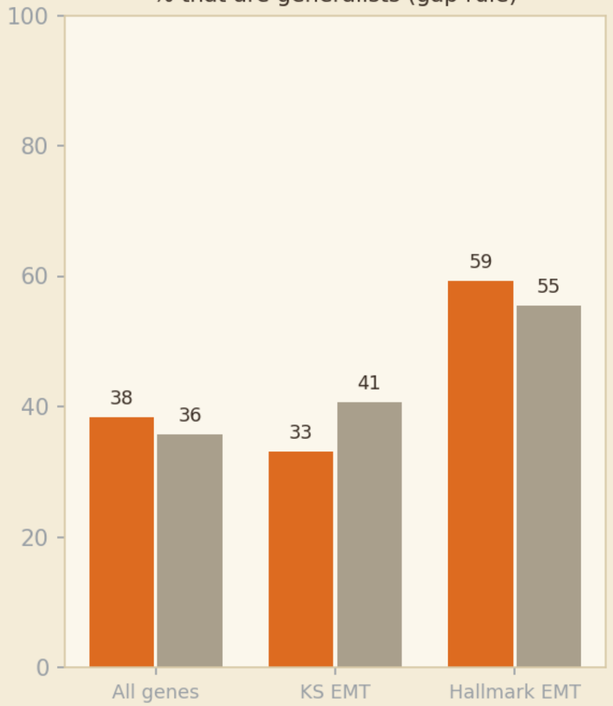
Pooled, all genes: corners mostly = cell lines
(cells coloured by cell line; k=3, p=0.0099)



Inside one cell line (A549, All genes): corners = programmes
(cells coloured by EMT state; k=3, p=0.01; 1 of 4 cell lines have a simplex)



Hybrid cells (orange) vs other cells (grey):
% that are generalists (gap rule)



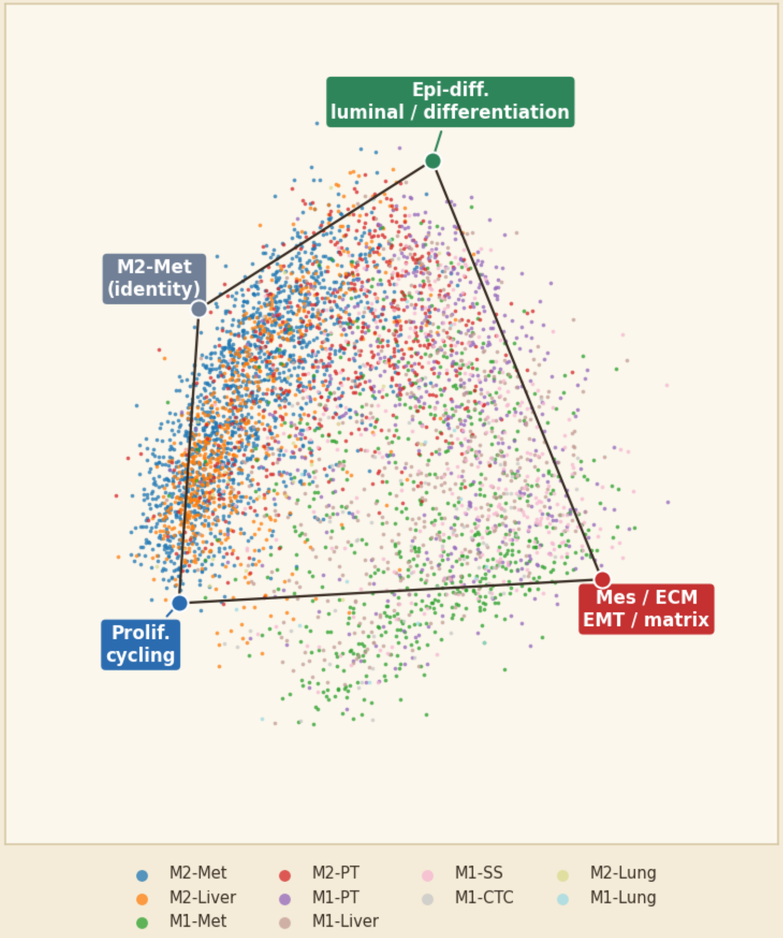
No difference between hybrids and others (median +0 points)

Significant within-cell line simplices: all genes 1/4, KS 1/4, Hallmark 3/4. Corner families across them: proliferation (3/5), epithelial differentiation (2/5), mesenchymal / matrix (4/5), metabolism & hypoxia (2/5), stress & immune signalling (2/5).

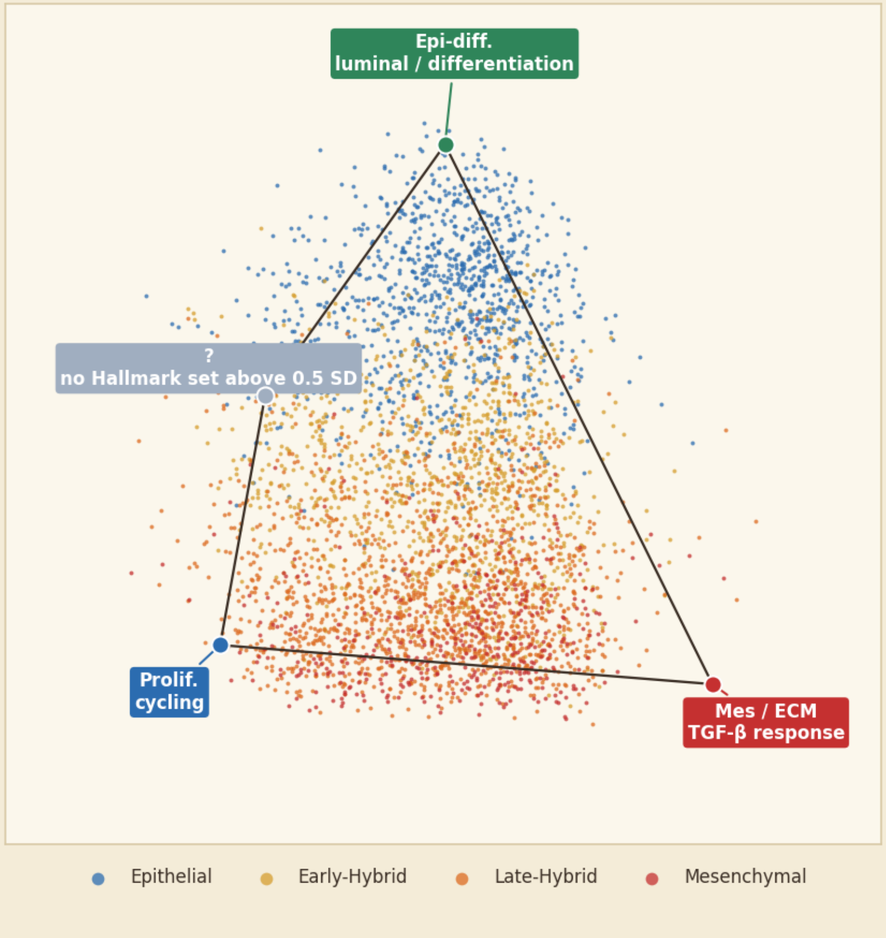
Mouse PDAC

Every PDAC library forms a simplex on all genes; its corners are proliferation and mesenchymal / matrix plus a metabolic or epithelial programme

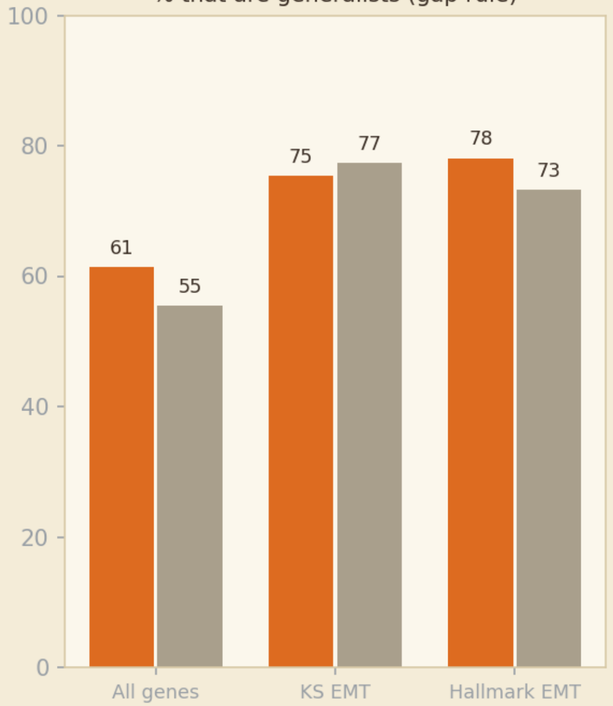
Pooled, all genes: partly libraries, partly programmes
(cells coloured by library; k=4, p=0.04)



Inside one library (M1, All genes): corners = programmes
(cells coloured by EMT state; k=4, p=0.005; 9 of 9 libraries have a simplex)



Hybrid cells (orange) vs other cells (grey):
% that are generalists (gap rule)

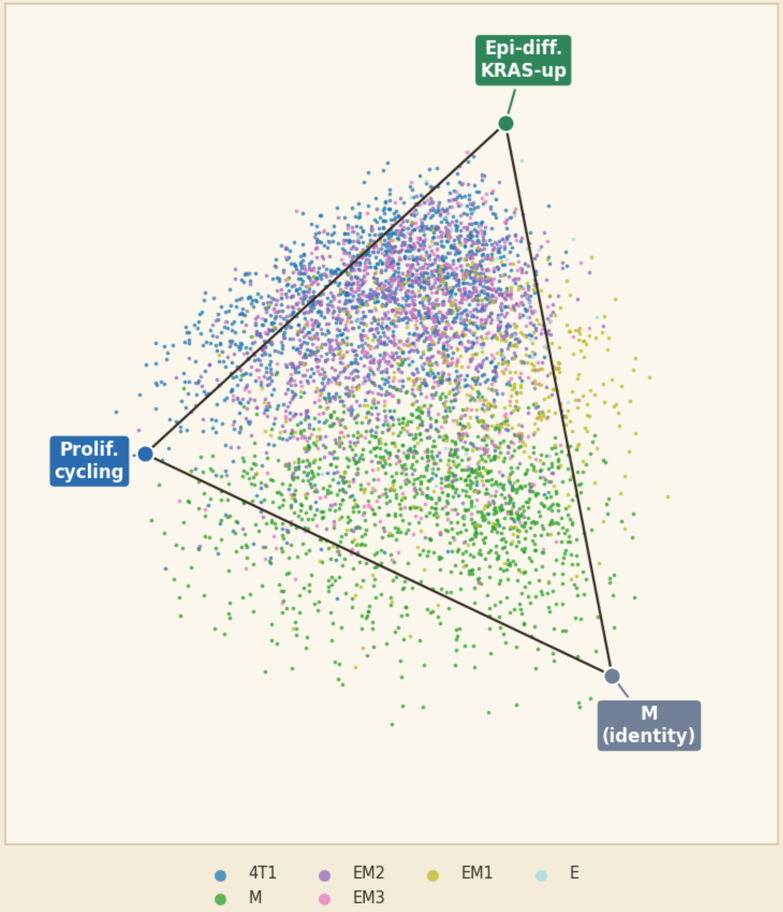


Hybrids lean slightly toward the interior (median +5 points)

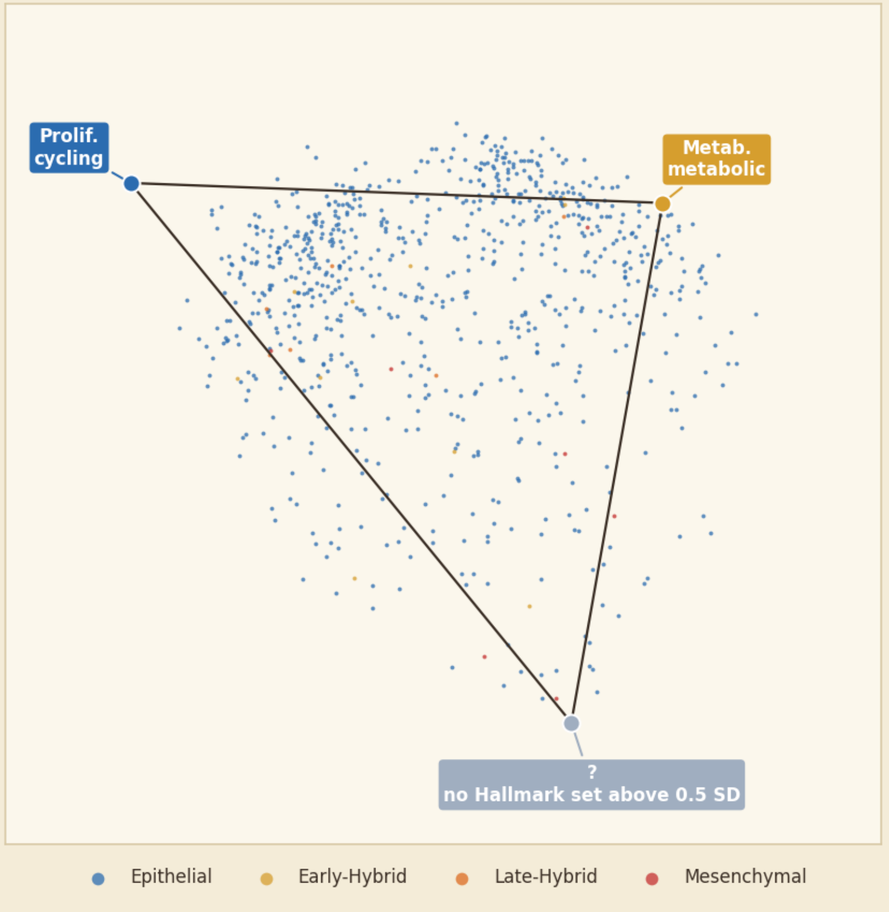
Significant within-library simplices: all genes 9/9, KS 3/9, Hallmark 4/9. Corner families across them: proliferation (12/16), epithelial differentiation (7/16), mesenchymal / matrix (13/16), metabolism & hypoxia (13/16), stress & immune signalling (1/16).

4T1 clones show almost no task structure: the pooled shape is mostly clone identity and single clones rarely form a simplex

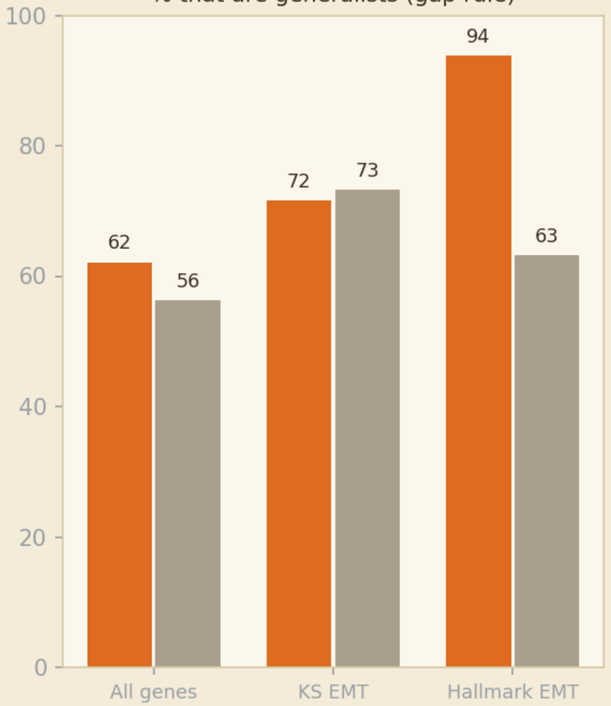
Pooled, all genes: partly clones, partly programmes
(cells coloured by clone; k=3, p=0.01)



Inside one clone (E, All genes): corners = programmes
(cells coloured by EMT state; k=3, p=0.01; 1 of 6 clones have a simplex)



Hybrid cells (orange) vs other cells (grey):
% that are generalists (gap rule)

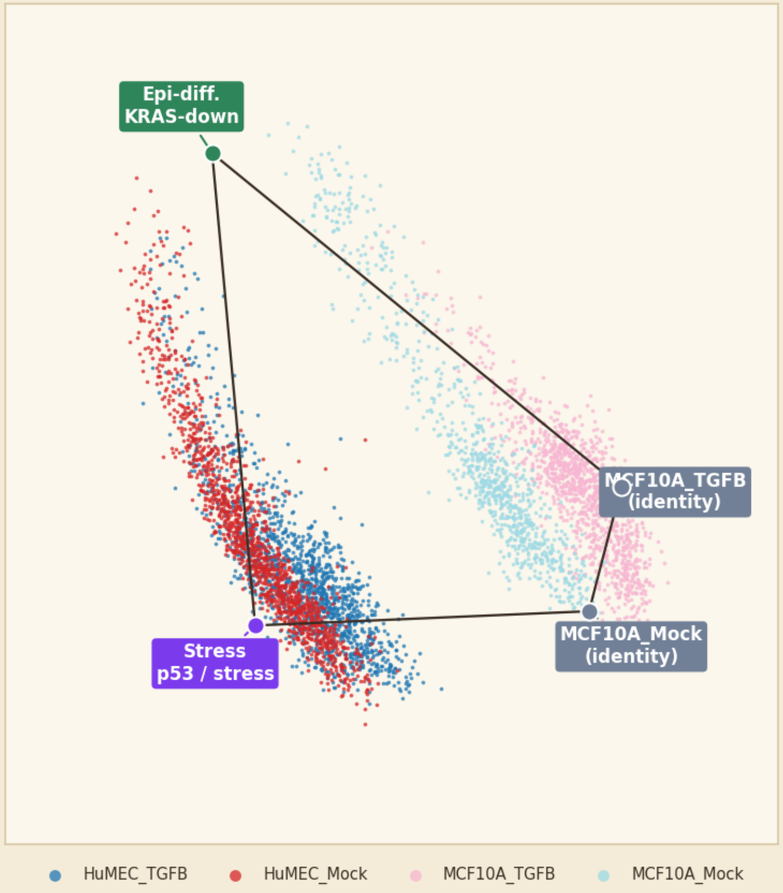


No difference between hybrids and others (median +2 points)

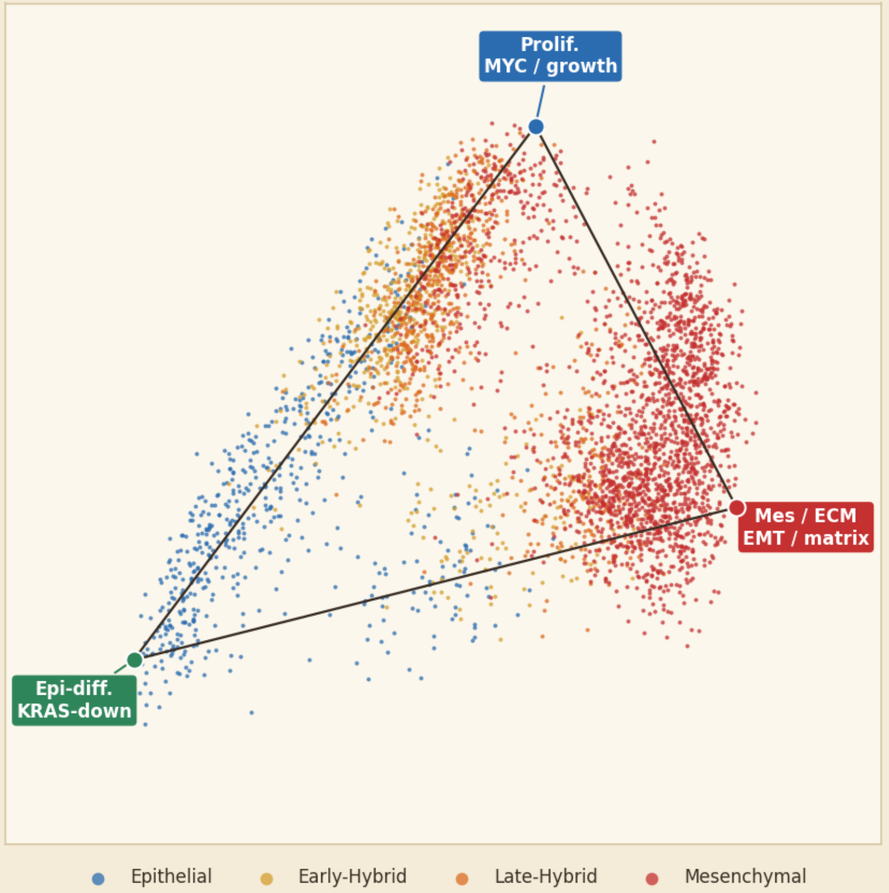
Significant within-clone simplices: all genes 1/6, KS 1/6, Hallmark 1/6. Corner families across them: proliferation (1/3), epithelial differentiation (2/3), mesenchymal / matrix (2/3), metabolism & hypoxia (1/3), stress & immune signalling (2/3).

MCF10A / HuMEC show the clearest trade-off: epithelial differentiation vs mesenchymal / matrix vs proliferation in every sample

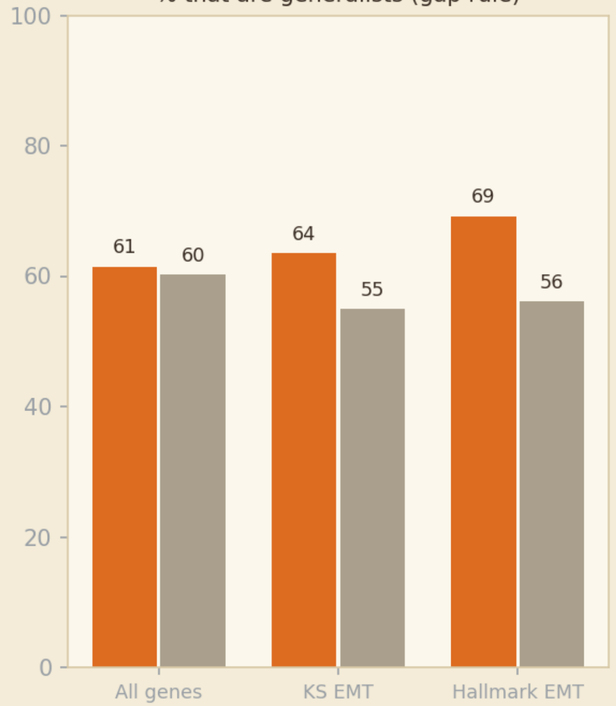
Pooled, all genes: partly samples, partly programmes
(cells coloured by sample; k=4, p=0.0099)



Inside one sample (MCF10A, All genes): corners = programmes
(cells coloured by EMT state; k=3, p=0.005; 6 of 6 samples have a simplex)



Hybrid cells (orange) vs other cells (grey):
% that are generalists (gap rule)



Hybrids lean slightly toward the interior (median +6 points)

Significant within-sample simplices: all genes 6/6, KS 4/6, Hallmark 5/6. Corner families across them: proliferation (8/15), epithelial differentiation (15/15), mesenchymal / matrix (11/15), metabolism & hypoxia (5/15), stress & immune signalling (6/15).